Kazi Ahsan Ahmed Akaid

Mechanical Engineering | Machine Learning | Prognostics & Health Management

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RESEARCH SUMMARY

Machine learning for reliability and manufacturing, with a focus on group-held-out evaluation, uncertainty quantification, and explanation agreement.

Interests: Prognostics and Health Management (PHM) | Remaining Useful Life (RUL) Prediction | Predictive Maintenance | Uncertainty Quantification | Explainable AI | Intelligent Manufacturing

EDUCATION

Chittagong University of Engineering & Technology (CUET) Chattogram, Bangladesh
Bachelor of Science in Mechanical Engineering Feb. 2022 – Jul. 2026
CGPA: **3.55/4.00** Final four-semester average: **3.80/4.00**
Undergraduate Thesis: *Predictive Modeling of Nanocellulose Characteristics from Acid Hydrolysis Process Parameters Using Machine Learning*

PUBLICATIONS & MANUSCRIPTS

Journal Manuscript

- **Akaid, K. A. A.**, and Fariha, F. "Calibration-Source Sensitivity and Warning Timing in Cutter-Held-Out Milling Tool-Wear Prediction." Under review at *Machining Science and Technology* (Submission ID 267468082).

Conference Papers

- **Akaid, K. A. A.**, and Rabbi, M. S. "Predictive Modeling of Nanocellulose Characteristics from Acid Hydrolysis Process Parameters Using Machine Learning." Submitted to ICMIME 2026, Rajshahi University of Engineering & Technology (RUET).
- **Akaid, K. A. A.** "Turbofan Engine Remaining Useful Life Prediction: Stacking Ensembles, Explanation Agreement, and Feature Pruning." Submitted to ICMIEE 2026, Khulna University of Engineering & Technology.

RESEARCH EXPERIENCE

Independent Researcher — Tool-Wear Prediction and Uncertainty Quantification 2026
Manuscript under review at Machining Science and Technology; PHM2010 and nine-tool milling datasets

- Showed that changing only the calibration cutter, with the fitted model and test predictions held fixed, moved nominal-90% interval coverage from 14.9% to 84.2% (Random Forest) and from 42.7% to 85.7% (CNN–BiLSTM), at the cost of much wider intervals (RF: 9.5 $\rightarrow$ 84.5 μm). The interleaved split behind this result was chosen after inspecting test outcomes, and the paper discloses this.
- Cutter-held-out MAE across C1/C4/C6 under a chronological split: 22.8 μm (RF) and 31.3 μm (CNN–BiLSTM). Under the interleaved split, a sensor-free cut-index control (21.1 μm) matched the CNN–BiLSTM (20.6 μm), so MAE alone does not show transferable wear information.
- At a 150 μm threshold, per-cut ROC-AUC exceeded 0.998, yet the first warning on cutter C6 came 29–39 cuts after the crossing. Ensemble disagreement did not track error (Spearman -0.07 to -0.14).
- Evaluated the interval procedure on an independent nine-tool milling dataset: 4.61 μm cutter-macro MAE and 97.3% coverage at nominal 90%.
- Code: github.com/aakaid010/phm2010-tool-wear-reliability.

Independent Researcher — Prognostics and Explainable AI 2025 – 2026
NASA C-MAPSS FD001/FD003; conference paper submitted to ICMIEE 2026

- Built a leakage-aware pipeline stacking Random Forest, XGBoost, LightGBM, and CatBoost with a Ridge meta-learner, using engine-grouped cross-validation. The best single learners reached test RMSE 17.57 (FD001) and 19.39 (FD003), and stacking did not beat them.
- Introduced cross-model SHAP explanation-agreement metrics (feature, rank, pairwise-rank agreement and Spearman correlation); mean pairwise-rank agreement of 0.87 (FD001) and 0.84 (FD003) explained why stacking added little.
- Leakage-free SHAP pruning with nested GroupKFold: on FD001, 5 of 51 features gave RMSE 17.30 versus 17.57 for the full set; FD003 needed about 30 features.
- Code: github.com/aakaid010/RUL-Prediction-using-Nasa-C-MAPSS-DATA.

Research Team Member — ELITE Research Lab Oct. 2026 – Present
"Garbage in, Bottleneck Out? How Machine-State Data Defects Mislead Data-Driven Throughput-Bottleneck Detection" (protocol stage; experiments not yet run)

- Contributing to a ground-truth benchmark of simulated serial and merging production lines whose bottlenecks are defined by throughput sensitivity, with an injection protocol for six machine-state log defects (timestamp resolution, unlogged short stops, state-vocabulary collapse, downtime misattribution, data gaps, clock offsets).
- Planned evaluation of six detection methods on accuracy, rank agreement, throughput regret, and stability, leading to a data-quality specification tested on held-out line configurations.

Undergraduate Thesis Researcher — ML for Nanocellulose Properties 2025 – 2026
Department of Mechanical Engineering, CUET; Supervisor: Dr. M. S. Rabbi Chattogram, Bangladesh

- Compiled 341 observations from 96 publications linking acid concentration, temperature, time, and feedstock to cellulose nanocrystal (CNC) crystallinity and length; compared seven regressors with paper-grouped splits and cross-validation.
- Random-split R^2 of 0.71 (crystallinity) and 0.86 (length) fell to negative (best -0.30) and near zero (best 0.05) under paper-grouped evaluation. The pipelines differ in more than the split, so this is reported as protocol sensitivity, not a causal leakage estimate.
- SHAP and impurity rankings agreed (Spearman 0.975 and 0.976). 90% conformal intervals reached 100% and 95.8% coverage but were too wide to be useful.
- Code: github.com/aakaid010/NanoCNC-Predictor.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB
- **Machine Learning:** scikit-learn, Random Forest, Gradient Boosting, SVR, XGBoost, LightGBM, CatBoost, Stacking Ensembles, CNN, BiLSTM
- **Explainable AI:** SHAP, Grad-CAM, Feature-Importance Analysis, Cross-Model Explanation-Agreement Metrics
- **Uncertainty Quantification:** Conformal Prediction, Prediction Intervals, Ensemble Disagreement
- **Data Science:** NumPy, pandas, Matplotlib, Feature Engineering, Feature Selection, Cross-Validation, Hyperparameter Optimization, Model Evaluation
- **PHM / Manufacturing:** NASA C-MAPSS, IMS, Paderborn, PHM2010, RUL Prediction, Tool-Wear Prediction, Predictive Maintenance
- **Engineering / Software:** SolidWorks, Flask, Git/GitHub, LaTeX
- **Research Platforms:** Google Colab, Kaggle

INDUSTRIAL EXPOSURE

Bangladesh Industrial Technical Assistance Center (BITAC) — Industrial Attachment Chattogram, Bangladesh

- Exposure to conventional and non-conventional machining, CNC machinery, welding, heat treatment, foundry, machine maintenance, and inspection and testing.

S. Alam Cold Rolled Steels Limited — Industrial Visit Chattogram, Bangladesh

- Observed cold-rolled steel production and steel-processing operations.

AWARDS & LEADERSHIP

- **Dean's List**, Department of Mechanical Engineering, CUET 3 semesters
- **Organizing Secretary**, Robo Mechatronics Association (RMA), CUET
  - Coordinated student events and activities for the university's robotics and mechatronics association.
- **General Member**, ASME Student Chapter, CUET

RELEVANT COURSEWORK

Statistical Analysis | Numerical Methods in Engineering | Measurement & Quality Control | Mechanics of Machinery & Vibration | Engineering Metallurgy

ENGLISH PROFICIENCY

IELTS Academic: Overall Band 7.5
Listening: 8.5 | Reading: 7.5 | Writing: 6.5 | Speaking: 7.0

REFERENCES

Dr. Md. Sanaul Rabbi
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Redoy Masum Meraz
Assistant Professor, Department of Mechanical Engineering
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Course Coordinator and Academic Advisor